

AgriCare: ML Powered Digital Solution for Agriculture

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Abstract - This research explores the use of advanced deep learning models to enhance the detection of plant diseases through image classification. The methodology incorporates both MobileNetV2 and a custom-designed convolutional neural network (CNN), each integrated with Efficient Channel Attention (ECA) mechanisms to improve feature sensitivity and classification reliability. Training and validation were conducted on an augmented dataset representing 33 categories of healthy and diseased leaf images. To enable well-informed agricultural recommendations, the framework also incorporates external factors such as environmental conditions and spatial data. The study demonstrates that compact, attention-enhanced neural models can provide accurate and scalable solutions for plant disease diagnosis in practical farming scenarios..

Keywords: Deep Learning, Crop Disease Detection, MobileNetV2, Crop Disease Classification, Efficient Channel Attention, Convolutional Neural Network (CNN), Image Classification, Leaf Image Dataset, Disease Diagnosis.

I. INTRODUCTION

Agriculture is the backbone of the Indian economy, employing a significant portion of the population. However, farmers face persistent challenges such as inadequate irrigation, fragmented land holdings, lack of access to modern technology, and low market prices. These factors contribute to financial instability and hinder agricultural growth.

This project aims to develop a digital platform tailored to farmers, equipping them with essential knowledge, real-time

market insights, and access to policy updates. By integrating technology, education, and government support, this initiative seeks to bridge the information gap, empower farmers, and promote sustainable agricultural development. The platform will provide guidance on modern farming techniques, government schemes, best practices for crop cultivation, and market intelligence, ensuring that farmers can make informed decisions to improve productivity and profitability.

II. LITERATURE REVIEW

Traditional methods for detecting plant diseases and making agricultural decisions often relied on manual inspection or basic machine learning algorithms, which lacked scalability and accuracy under real-world field conditions. With the introduction of deep learning—especially convolutional neural networks (CNNs)—image-based classification for crop disease detection has seen a marked improvement in both precision and reliability ^{[1][2]}

Among deep learning models, MobileNetV2 has gained traction due to its lightweight design, making it suitable for mobile deployment and use in low-resource environments ^[3]. When paired with Efficient Channel Attention (ECA) mechanisms, these networks can enhance their focus on critical visual features, improving their ability to distinguish between healthy and diseased areas in plant imagery ^[4]. Similarly, incorporating ECA into custom CNN models offers robustness and flexibility in environments with noise or data imbalance ^{[4][5]}.

Preprocessing techniques like resizing, normalization, and extensive image augmentation—such as flipping, rotation, and brightness alteration—help mimic field

variability and enhance model generalization while reducing overfitting risks.^[6] Techniques like label smoothing, dropout regularization, and validation callbacks further ensure stable model training and convergence.^[7]

All these innovations—ranging from attention-enhanced CNN architectures and intelligent preprocessing to imbalance mitigation and hyperparameter tuning—together contribute to a scalable, efficient, and practical digital agriculture platform.^[10]

III. METHODOLOGY

The crop disease detection system is built on two deep learning models :

- (i) a MobileNetV2 architecture integrated with Efficient Channel Attention (ECA),^{[3][4]} and
- (ii) a custom Convolutional Neural Network (CNN) also enhanced with ECA blocks.

Both models are trained on an augmented dataset of plant leaf images covering 33 disease classes^[13], with the aim of learning to distinguish between healthy and diseased conditions across a wide range of crops.

To simulate real-world variability in leaf appearance, images are adjusted in size and brightness to simulate natural conditions and improve learning.^[5] These transformations improve model generalization and robustness during deployment.

Efficient Channel Attention (ECA)

Fig 1. illustrates the Efficient Channel Attention (ECA) mechanism that improves image classification models by selectively emphasizing important feature channels without significantly increasing computational cost. Unlike traditional attention modules that rely on fully connected layers for channel interaction, ECA introduces efficient strategy by combining global average pooling with a lightweight 1D convolution. First, the spatial dimensions of the input feature map are globally averaged to capture channel-wise statistics, producing a compact vector representing the importance of each channel. A lightweight 1D convolution, whose kernel size adjusts with the number of channels and learns localized relationships across channels. A sigmoid activation function scales these values to produce attention weights, which are then reshaped and multiplied back with the original input. This multiplication applies the learned attention to each channel, highlighting informative features while suppressing less relevant ones. By refining the feature representation in this way, ECA selectively amplifies key feature channels using global average pooling and lightweight convolutions, improving accuracy with minimal increase in model size or latency^[4].

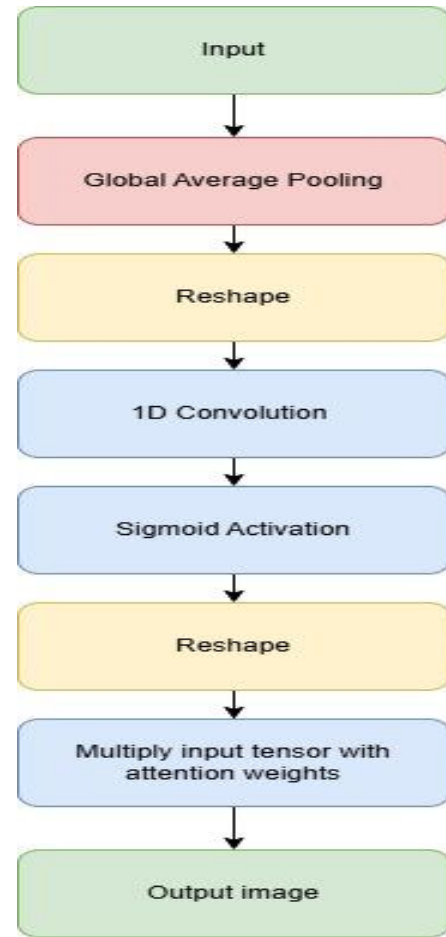


Fig. 1: ECA layout

MobileNetV2 + ECA

Fig 2. illustrates the first model which utilizes the MobileNetV2 backbone^[3] for its efficiency and suitability for mobile deployment. Transfer learning is applied using pre-trained weights from ImageNet^[11], with selective layer fine-tuning to adapt to the plant disease classification task. The ECA module is used to improve channel-wise feature selection.

Custom CNN + ECA

Fig 3. illustrates the second model which is a custom CNN built from scratch, featuring convolutional layers with LeakyReLU activations, batch normalization, and ECA modules after each block. It uses pooling, dropout, and dense layers for feature abstraction and classification. Unlike the MobileNet-based model, it learns directly from scratch on the plant dataset.

Together, these two attention-enhanced models provide very accurate disease classification, contributing to a scalable system for plant health monitoring.

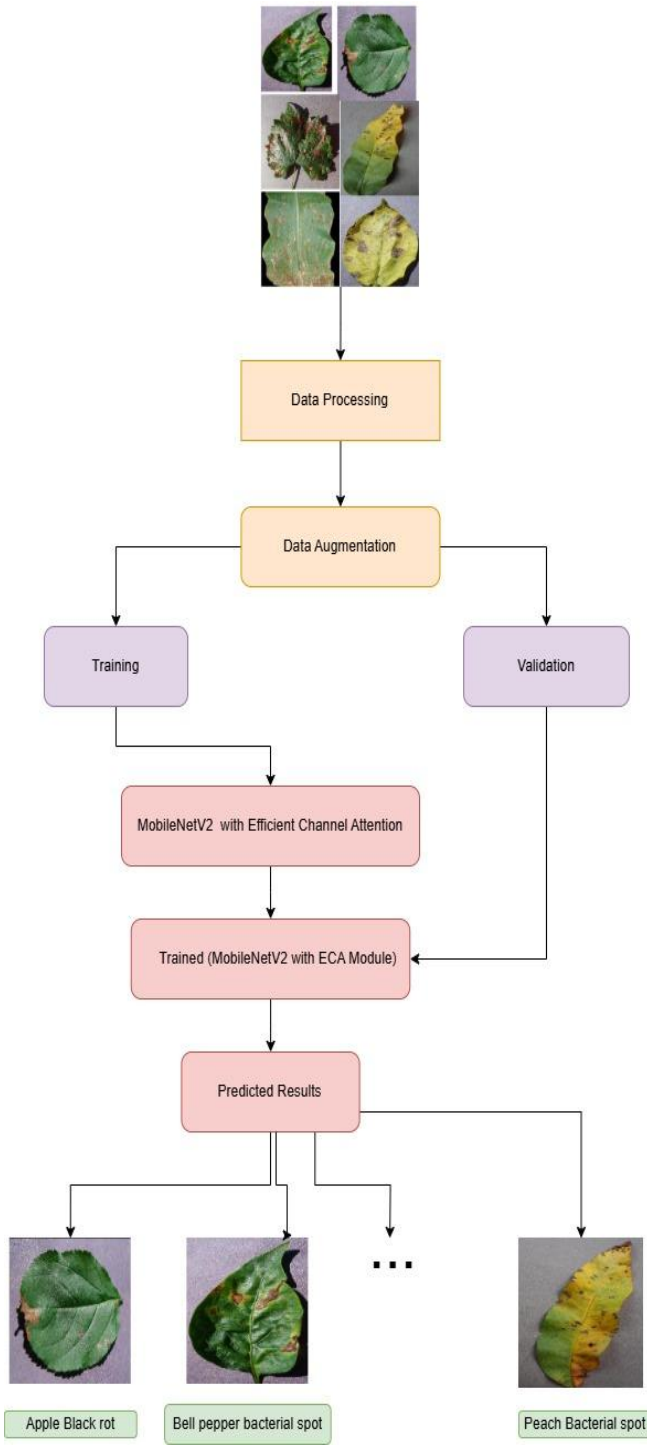


Fig 2: Model architecture (MobileNetV2)

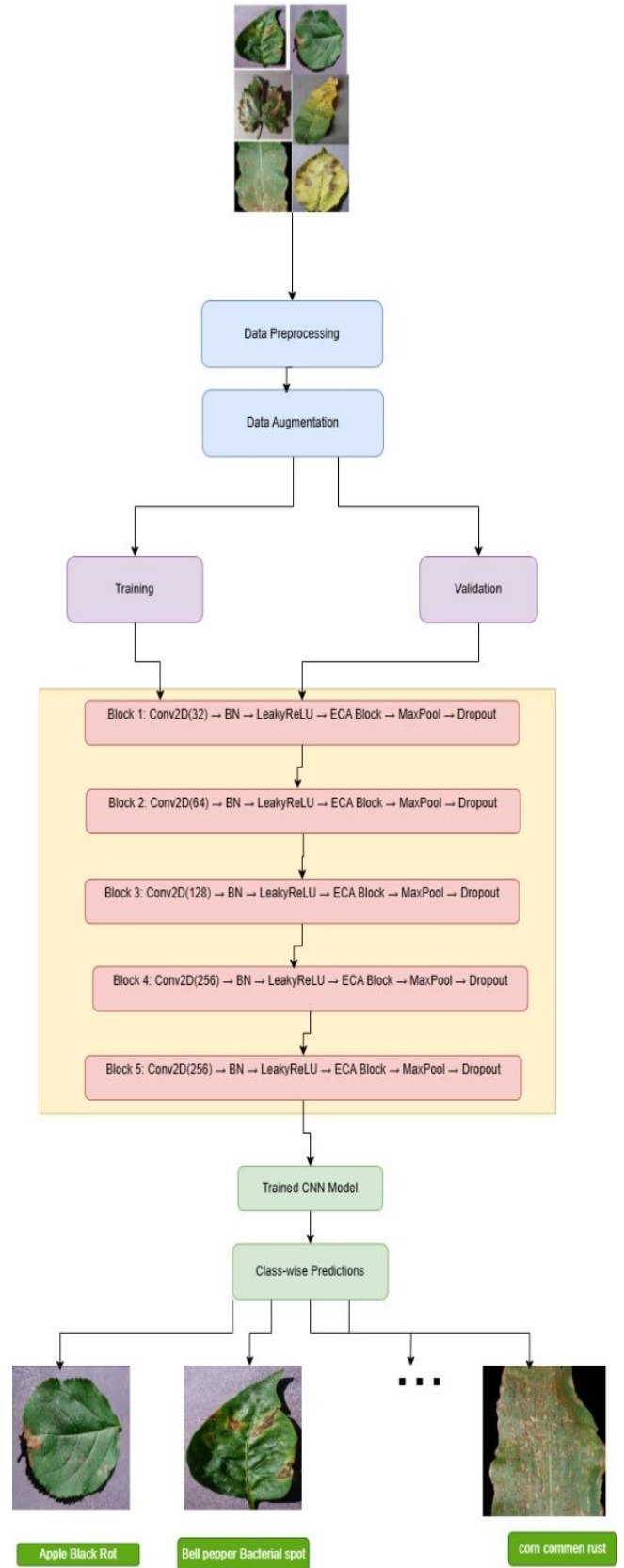


Fig. 3: Model Architecture (Custom CNN)

A. Dataset Description

The plant disease classification dataset includes 33 distinct classes^[13] encompassing both healthy and infected leaves from crops such as Apple, Bell Pepper, Cedar, Citrus,

Corn, Grape, Peach, Potato, and Tomato, Fig 4 shows some sample image from the dataset. Each class varies in size — for instance, Tomato Yellow Leaf Curl Virus has 7,877 images, while Citrus Healthy has just 87. To maintain diversity and fairness across the dataset, care was taken to retain a proportional representation of each class. The dataset is split into training and validation sets using a stratified sampling approach, with 80% of the data used for training and the remaining 20% reserved for validation. This ensures the validation set contains representative samples from all classes, allowing for more accurate and consistent performance evaluation. The partitioning strategy supports the development of a reliable classification model capable of generalizing well across different plant diseases.

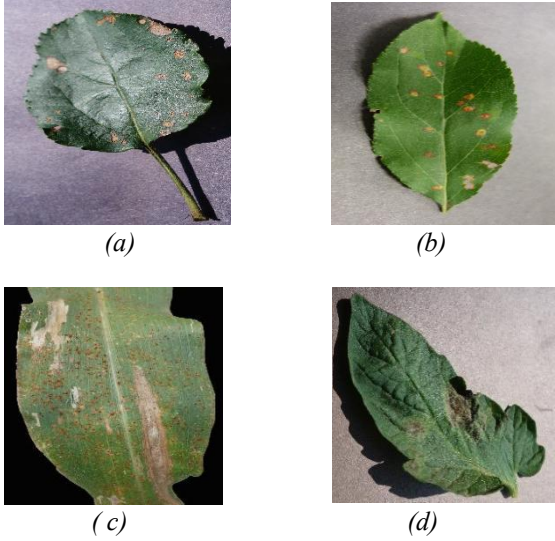


Fig. 4: Sample image of dataset (a) Apple Black Rot, (b) Cedar Apple Rust, (c) Corn Common Rust, (d) Tomato Late Blight

B. Data Processing and Augmentation

Before initiating the training process, all crop disease images undergo systematic preprocessing and augmentation to improve the model's ability to generalize across diverse agricultural conditions. This stage ensures that input images are uniformly resized, pixel values are normalized, and data is formatted to align with the model's input requirements. This simplifies the input pipeline and ensures stable and efficient learning during training. Augmentation techniques are then applied to artificially expand the dataset by introducing controlled randomness through spatial and photometric transformations. These include image flips, rotations, shifts, zooms, and brightness adjustments. By doing so, the model is exposed to a wide range of possible field conditions, plant orientations, and lighting scenarios—thereby reducing overfitting and enhancing robustness to real-world variations. Such transformations play a critical role in helping the model capture the underlying patterns associated with different crop diseases, even when they appear in unfamiliar forms. As a result, the model becomes more accurate, resilient, and reliable in practical applications, especially when deployed for large-scale agricultural diagnostics under variable environmental conditions.

C. Model Architecture

a) MobileNetV2: The classification model developed in this study utilizes the MobileNetV2 architecture, which is known for its efficiency and suitability for mobile and embedded vision applications. To improve the feature representation, an ECA module is integrated into the architecture as illustrated in Fig 5. The model begins with a MobileNetV2 backbone pre-trained on ImageNet, which allows reuse of general visual features through transfer learning. Transfer learning is employed by freezing the base layers during the initial training phase, followed by selective unfreezing and fine-tuning in subsequent epochs.

The model architecture comprises the MobileNetV2 backbone without the top classification layers, followed by a custom head. This head includes a 2D convolutional layer to increase channel depth, batch normalization, and a ReLU activation. The output is passed through the ECA attention module, which adjust channel importance by capturing localized inter-channel dependencies using computationally light 1D convolutions^[4].

The attention-refined feature maps are further processed using global average pooling, dropout, and dense layers to reduce overfitting and enhance generalization. The final layer is a softmax classifier that outputs probabilities across multiple classes. The model is trained using categorical cross-entropy loss with label smoothing and optimized with the Adam optimizer^{[14][15]}. The architecture is designed to extract salient features from images, enabling accurate classification of diseases into predefined categories.

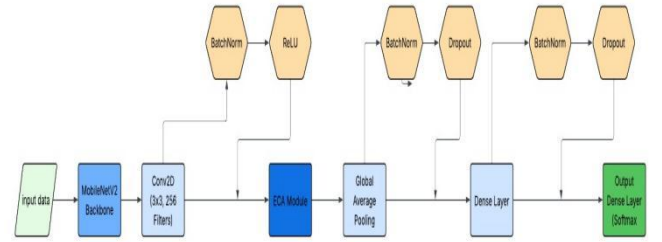


Fig. 5: MobileNetV2 architecture with ECA

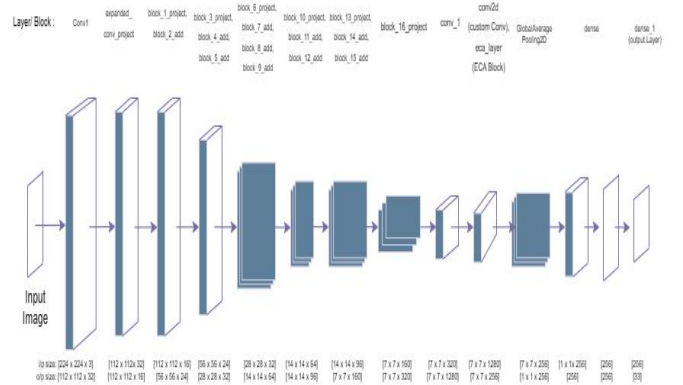


Fig. 6: Image Feature Map Evolution in MobileNetV2 + ECA Architecture

b) Custom CNN : The classification model developed in this study is a custom Convolutional Neural Network (CNN) enhanced with an ECA mechanism illustrated in Fig

7. This architecture is designed specifically for the multi-class classification of plant diseases from RGB images.

The model begins with an input layer that processes the image data, followed by a series of convolutional layers. Each convolutional block includes batch normalization and LeakyReLU activation to stabilize and enhance feature extraction. To improve the model's ability to focus on the most informative channels, the ECA module is integrated after each block. This module adaptively recalibrates channel-wise feature responses without introducing significant computational overhead.

Spatial downsampling is performed using pooling operations, helping preserve essential semantic features while reducing computational load, while retaining key features. Dropout layers are incorporated after each block to prevent overfitting and ensure generalization.

The feature maps are then aggregated using a global average pooling layer, followed by a fully connected dense layer for high-level feature abstraction. The final classification is performed by a softmax layer, which outputs probabilities across all target classes.

For training, the model uses a smoothed categorical cross-entropy loss function and is optimized with the Adam optimizer^[15] to ensure stable and efficient convergence. Data augmentation techniques such as flipping, rotation, zooming, and brightness adjustment are applied during training to improve robustness and reduce overfitting.

This architecture effectively combines convolutional feature extraction with attention mechanisms to achieve accurate and reliable classification of plant disease categories.

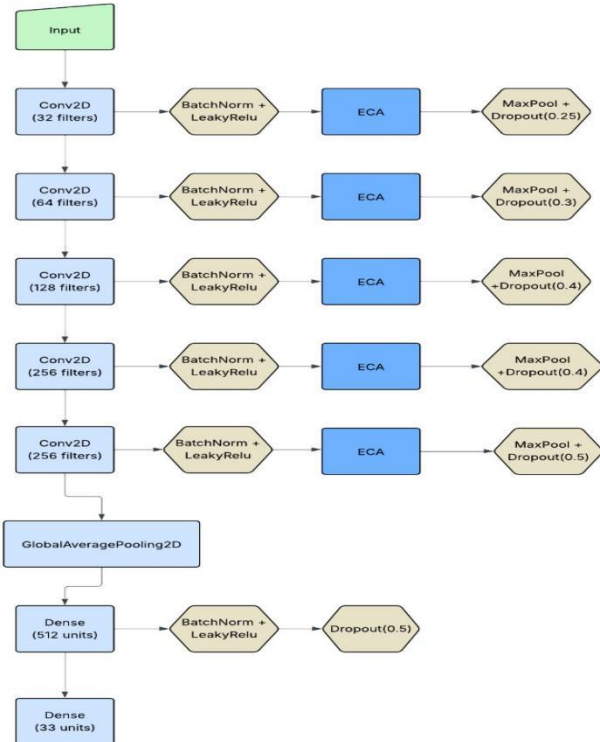


Fig. 7: Custom CNN Architecture with ECA

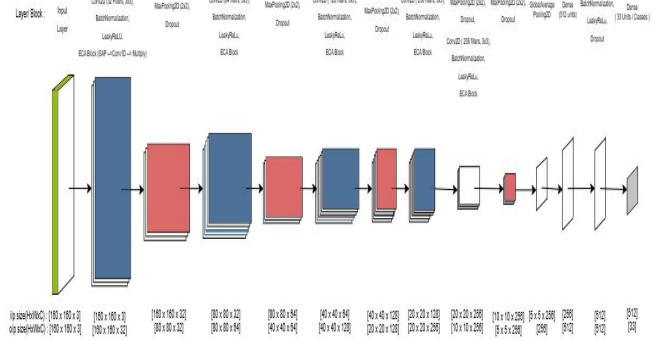


Fig. 8: Image Feature Map Evolution in Custom CNN + ECA Architecture

IV. RESULTS AND DISCUSSION

Both the MobileNetV2+ECA and custom CNN+ECA architectures demonstrated strong performance in multi-class crop disease classification across 33 categories, as reflected in key evaluation metrics and training trends. Training trends are shown in the accuracy and loss plots (Fig. 9 and Fig. 10), while performance metrics in Table 1 and Table 2 highlight the classification effectiveness across all 33 classes.

A. Accuracy and Loss Plots

Fig. 9 and Fig. 10 depict the progression of model accuracy and loss, respectively, over the course of training epochs. These visualizations help us assess how well the model is learning to classify plant disease images by comparing training and validation performance.

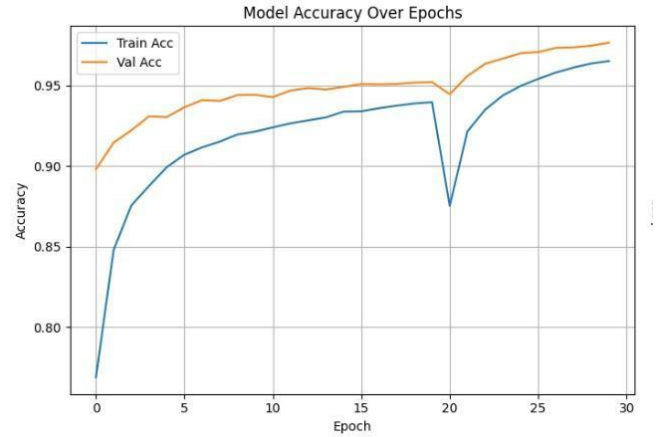


Fig. 9: Training and Validation Accuracy for MobileNetV2 with ECA

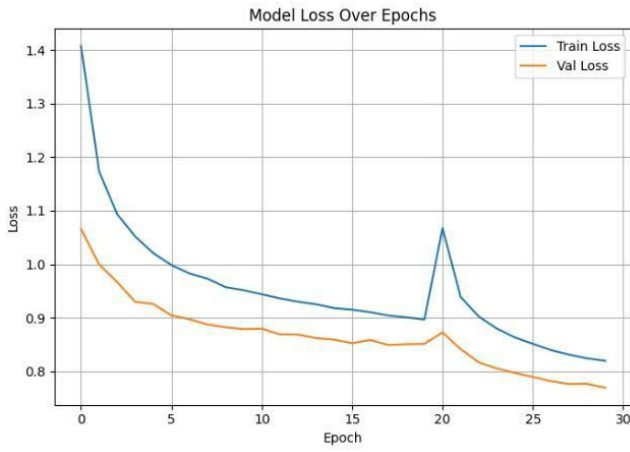


Fig. 10: Training and Validation Loss for MobileNetV2 with ECA

Fig. 9 shows the evolution of training and validation accuracy over 30 epochs. Initially, the training accuracy starts at approximately 77%, while the validation accuracy is slightly higher at 90%. Throughout the epochs, both metrics exhibit a general upward trend, suggesting consistent learning. By the final epoch, the training accuracy reaches around 96.3%, with validation accuracy peaking at approximately 97.5%. Despite a small dip around epoch 20 in training accuracy, the validation accuracy continues to improve, indicating strong generalisation ability. This performance pattern implies that the model successfully captures relevant features from the input data and is robust to unseen samples.

Fig. 10 illustrates the corresponding training and validation loss trends. The training loss begins at around 1.4, which is slightly higher than the initial validation loss of approximately 1.07. As training progresses, both losses decline steadily, demonstrating effective error minimization. A temporary spike at epoch 20 in training loss is observed but is quickly corrected in subsequent epochs. The downward trend in loss values is a clear sign of model convergence and successful learning. The consistent gap between training and validation loss, without significant divergence, suggests minimal overfitting.

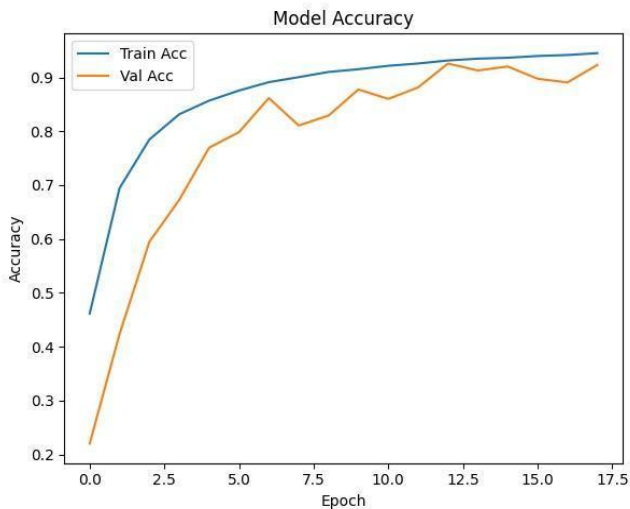


Fig. 11: Training and Validation Accuracy for Custom CNN with ECA

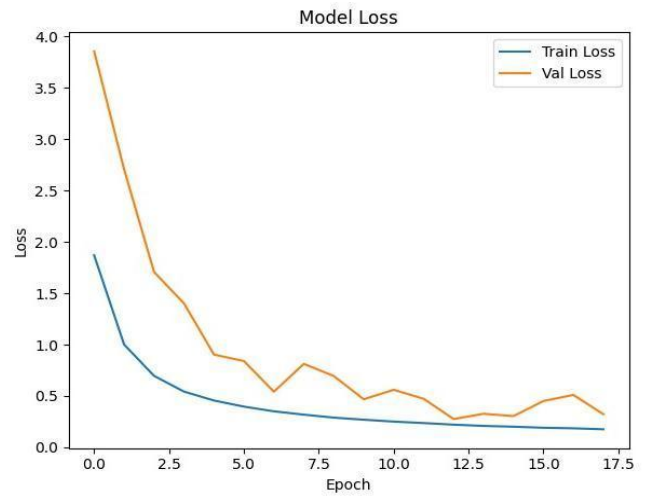


Fig. 12: Training and Validation Loss for Custom CNN with ECA

Fig. 11 and Fig. 12 represent another model's training trajectory, based on a CNN with fewer epochs (up to 17). In Fig. 11, the custom CNN began with relatively low performance, typical of models trained from scratch. However, both training and validation accuracies improved consistently, reaching above 90% by the end of 17 epochs, confirming the model's learning ability despite a limited training cycle. Minor fluctuations in validation accuracy are observed around epochs 6 and 14, yet the overall curve confirms effective generalisation.

Fig. 12 shows the corresponding loss reduction. The training loss begins at 1.88, while the validation loss starts high at around 3.88. Both losses exhibit a sharp decline, with training loss eventually approaching near-zero and validation loss dropping significantly to under 0.4 by the final epoch. This clear reduction in both losses confirms the model's efficiency in learning from training data and transferring that understanding to the validation set.

B. Performance Parameters

To assess the classification capability of the proposed models, we evaluated standard metrics including Precision, Recall, and F1-Score across 33 crop disease classes. The performance of MobileNetV2+ECA and a custom CNN+ECA model is summarized in Tables 1 and 2, respectively. These metrics provide insight into how effectively each model handles multi-class classification, ensuring their applicability for real-time agricultural diagnostics.

TABLE 1 : Classification Report of MobileNetV2

<i>Class Name</i>	<i>Precision</i>	<i>Recall</i>	<i>F1-Score</i>
Apple Black rot	1.0000	0.9968	0.9984
Apple Healthy	0.9906	0.9917	0.9911
Apple Scab	0.9803	1.0000	0.9900
Bell pepper Bacterial spot	0.9617	0.9840	0.9727
Bell pepper Healthy	0.9721	0.9975	0.9846
Cedar apple rust	0.9921	0.9902	0.9912

<i>Class Name</i>	<i>Precision</i>	<i>Recall</i>	<i>F1-Score</i>
Citrus Black spot	0.3896	0.9677	0.5556
Citrus Healthy	0.7692	0.5882	0.6667
Citrus canker	0.9394	0.8493	0.8921
Citrus greening	0.9944	0.9196	0.9555
Corn Common rust	0.9946	0.9918	0.9932
Corn Gray leaf spot	0.9080	0.9771	0.9413
Corn Healthy	1.0000	1.0000	1.0000
Corn Northern Leaf Blight	0.9763	0.9295	0.9523
Grape Black Measles	0.9987	0.9974	0.9980
Grape Black rot	0.9944	0.9958	0.9951
Grape Healthy	0.9981	0.9961	0.9971
Grape Isariopsis Leaf Spot	1.0000	1.0000	1.0000
Peach Bacterial spot	0.9989	0.9978	0.9984
Peach Healthy	0.9774	0.9924	0.9848
Potato Early blight	0.9773	0.9773	0.9773
Potato Healthy	0.9938	0.9815	0.9876
Potato Late blight	0.9727	0.9602	0.9664
Tomato Bacterial spot	0.9855	0.9334	0.9587
Tomato Early blight	0.9752	0.9036	0.9380
Tomato Healthy	0.8902	0.9914	0.9381
Tomato Late blight	0.9668	0.9769	0.9718
Tomato Leaf Mold	0.9859	0.9291	0.9567
Tomato Mosaic virus	0.9868	0.9868	0.9868
Tomato Septoria leaf spot	0.9460	0.9425	0.9443
Tomato Spider mites	0.9775	0.8468	0.9074
Tomato Target Spot	0.8367	0.9593	0.8938
Tomato Yellow Leaf Curl Virus	0.9955	0.9930	0.9943

Overall Accuracy	—	—	0.9693
Macro Average	0.9493	0.9559	0.9479
Weighted Average	0.9730	0.9693	0.9702

TABLE 2 : Classification Report of custom CNN

<i>Class Name</i>	<i>Precision</i>	<i>Recall</i>	<i>F1-Score</i>
Apple Black rot	0.8584	0.9758	0.9133
Apple Healthy	0.4738	0.9634	0.6352
Apple Scab	0.9773	0.4659	0.6310
Bell pepper Bacterial spot	0.9804	0.5820	0.7304
Bell pepper Healthy	0.6995	0.8940	0.7849
Cedar apple rust	0.9798	0.9529	0.9662
Citrus Black spot	0.5000	0.0806	0.1389
Citrus Healthy	0.7273	0.4706	0.5714

<i>Class Name</i>	<i>Precision</i>	<i>Recall</i>	<i>F1-Score</i>
Citrus canker	0.7879	0.3562	0.4906
Citrus greening	0.8339	0.9810	0.9015
Corn Common rust	0.9815	0.9375	0.9590
Corn Gray leaf spot	0.8868	0.8952	0.8910
Corn Healthy	0.9406	1.0000	0.9694
Corn Northern Leaf Blight	0.8728	0.9676	0.9177
Grape Black Measles	0.9313	0.9868	0.9583
Grape Black rot	0.9864	0.6050	0.7500
Grape Healthy	0.8885	0.9691	0.9271
Grape Isariopsis Leaf Spot	0.8031	0.9736	0.8802
Peach Bacterial spot	0.7899	0.9913	0.8792
Peach Healthy	0.9797	0.7366	0.8410
Potato Early blight	0.9222	0.9405	0.9313
Potato Healthy	0.9037	0.3477	0.5022
Potato Late blight	0.8786	0.9148	0.8963
Tomato Bacterial spot	0.8474	0.8542	0.8508
Tomato Early blight	0.9769	0.4863	0.6494
Tomato Healthy	0.4158	0.9914	0.5859
Tomato Late blight	0.7030	0.8438	0.7670
Tomato Leaf Mold	0.9680	0.6706	0.7923
Tomato Mosaic virus	1.0000	0.4633	0.6332
Tomato Septoria leaf spot	0.8858	0.7115	0.7892
Tomato Spider mites	0.7988	0.6649	0.7257
Tomato Target Spot	1.0000	0.2035	0.3382
Tomato Yellow Leaf Curl Virus	0.9910	0.9092	0.9483

Overall Accuracy	—	—	0.8083
Macro Average	0.8536	0.7511	0.7620
Weighted Average	0.8635	0.8083	0.8020

Evaluation metrics like Overall Accuracy, Macro Average, and Weighted Average help us understand the performance of a multi-class classification model.

Precision, Recall, and F1 Score are used within both Macro and Weighted Averages to provide a deeper look at how the model behaves.

In **Macro Average**, the model calculates precision, recall, and F1 score for each class individually and then averages them equally. This helps us understand how well the model performs across all classes, especially less common ones.

In **Weighted Average**, the same scores are calculated per class, but more weight is given to classes with more samples. This gives a more accurate view of how the model performs on the entire dataset, especially if some classes appear more frequently.

C. Confusion Matrix

Fig. 13 illustrates the confusion matrix generated from the crop disease classification model based on MobileNetV2 enhanced with Efficient Channel Attention (ECA).

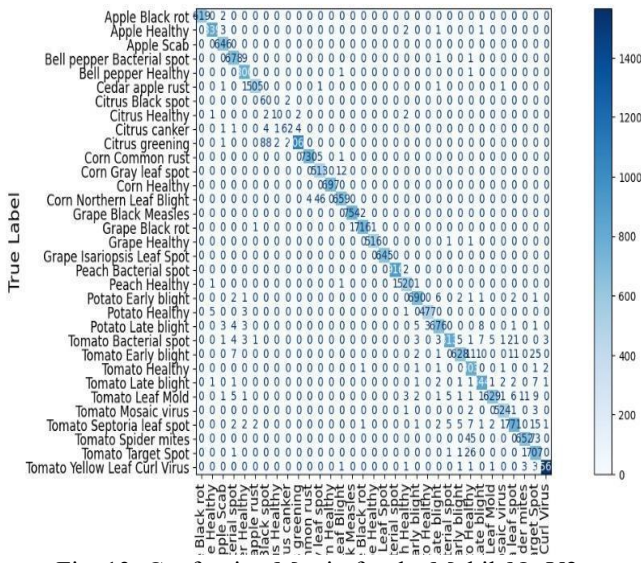


Fig. 13: Confusion Matrix for the MobileNetV2

The matrix summarizes the model's performance across 33 distinct classes of healthy and diseased plant leaves from multiple crops, including apple, grape, citrus, corn, potato, and tomato. Each row in the matrix represents the actual class labels, while each column represents the predicted class labels. A dense concentration of values along the diagonal signifies a high number of true positive classifications, suggesting that the model correctly predicts a significant proportion of instances for most classes. For example, the model demonstrates near-perfect classification in categories such as Tomato Leaf Mold, Cedar Apple Rust, and Tomato Septoria Leaf Spot, where misclassification rates are almost negligible. However, some degree of confusion is noticeable in visually similar classes—particularly between different diseases within the same crop family, such as various tomato leaf conditions (Early blight, Late blight, and Target Spot) and grape-related diseases like Black rot and Leaf blight. Additionally, healthy leaf classes tend to have a higher rate of misclassification, especially in crops like apple and tomato, which may be attributed to subtle visual differences or early-stage infections that resemble disease symptoms. The model's limitations are also influenced by class imbalance, as categories with fewer training samples tend to show reduced performance, indicating a need for improved data augmentation or rebalancing techniques. Overall, the confusion matrix reveals that the model is effective at distinguishing between most diseases while also highlighting areas where additional refinement is needed. To evaluate this performance quantitatively, several metrics are derived from the confusion matrix. Accuracy is defined as the ratio of correctly predicted observations to the total observations, given by the formula:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively. Precision, which measures the proportion of correctly predicted positive observations to the total predicted positives, is given by:

$$\text{Precision} = \frac{TP}{TP + FP}$$

and F1-score, the harmonic mean of precision and recall, provides a balance between the two and is defined as:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

These metrics together offer a comprehensive understanding of the model's strengths and weaknesses in multi-class classification tasks involving plant disease detection.

Fig. 14 illustrates the confusion matrix which represents the performance of the custom CNN model enhanced with Efficient Channel Attention (ECA) on a 33-class plant disease classification task.

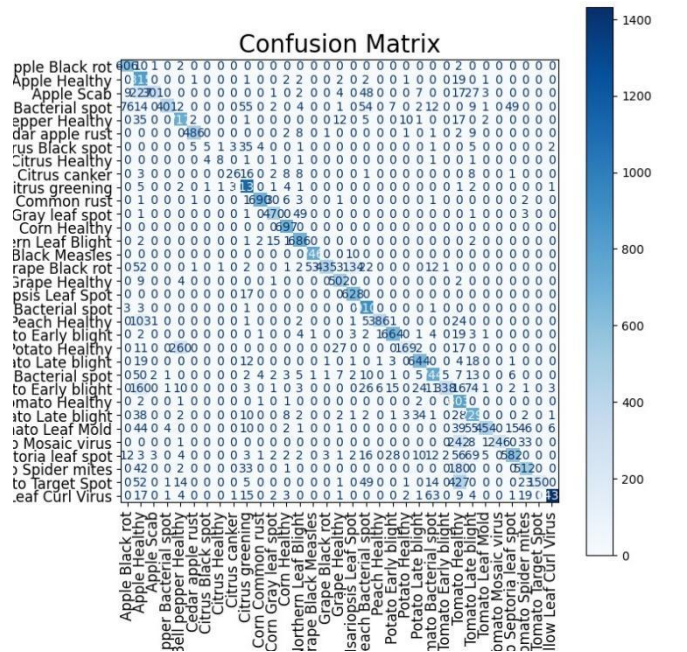


Fig. 14: Confusion Matrix for custom CNN model

The model demonstrates strong classification ability with most of the classes showing high true positive counts, as evident from the dense diagonal in the matrix. For example, the model correctly classified 1338 images of Tomato Late Blight, 1246 images of Tomato Septoria Leaf Spot, and 1273 samples of Apple Scab. Similarly, healthy classes like Corn Healthy and Potato Healthy also exhibit high true positive values of 866 and 692 respectively, indicating robust learning of distinguishing features. Despite the overall strong performance, a few confusions persist,

especially in categories with similar visual symptoms. For instance, Tomato Target Spot and Tomato Bacterial Spot share close symptom patterns, resulting in moderate misclassifications. Some healthy leaf categories, such as Apple Healthy and Bell Pepper Healthy, show lower precision, possibly due to the model mistaking early disease onset as healthy or vice versa. The overall model performance is promising, with a macro precision of approximately 0.80, recall of 0.80, and F1-score of 0.79, suggesting a well-balanced classification performance across all classes. These results underscore the effectiveness of the ECA-integrated CNN architecture in capturing complex visual patterns in plant disease imagery while also highlighting specific class pairs that may benefit from targeted augmentation or refined feature extraction.

D. Model Interpretability with Grad-CAM

To enhance the interpretability of the crop disease classification models, we employed **Grad-CAM (Gradient-weighted Class Activation Mapping)**,^[11] which provides a visualization interpretation by generating heatmaps over areas in input images that significantly influence the model’s classification output. This was applied to both the MobileNetV2 + ECA and Custom CNN + ECA architectures. By overlaying the activation heatmaps on original leaf images, we were able to observe whether the models were attending to the disease-affected regions.

As illustrated in Fig. 15 (a–d), the Grad-CAM outputs demonstrate the models’ ability to localize disease-relevant features. For instance, in cases of *Apple Black Rot* and *Potato Late Blight*, the highlighted zones in the Grad-CAM images align with visibly affected areas on the leaves. In the MobileNetV2-based model, heatmaps were more concentrated and centered over the lesion areas, indicating more precise feature localization. Meanwhile, the custom CNN also performed well, though some heatmaps were slightly broader or diffused, suggesting that it sometimes captures surrounding contextual features in addition to the primary symptoms.

Moreover, comparison between correctly classified and misclassified samples helped identify situations where the models’ attention diverged from the actual diseased regions—often due to ambiguous patterns or overlapping symptoms. For example, in some custom CNN predictions, the Grad-CAM maps revealed attention around healthy leaf edges or background textures, potentially leading to incorrect classifications. This interpretability step is critical, as it ensures the models are not relying on irrelevant cues and builds trust in their predictions by confirming that their decisions are grounded in biologically meaningful evidence.

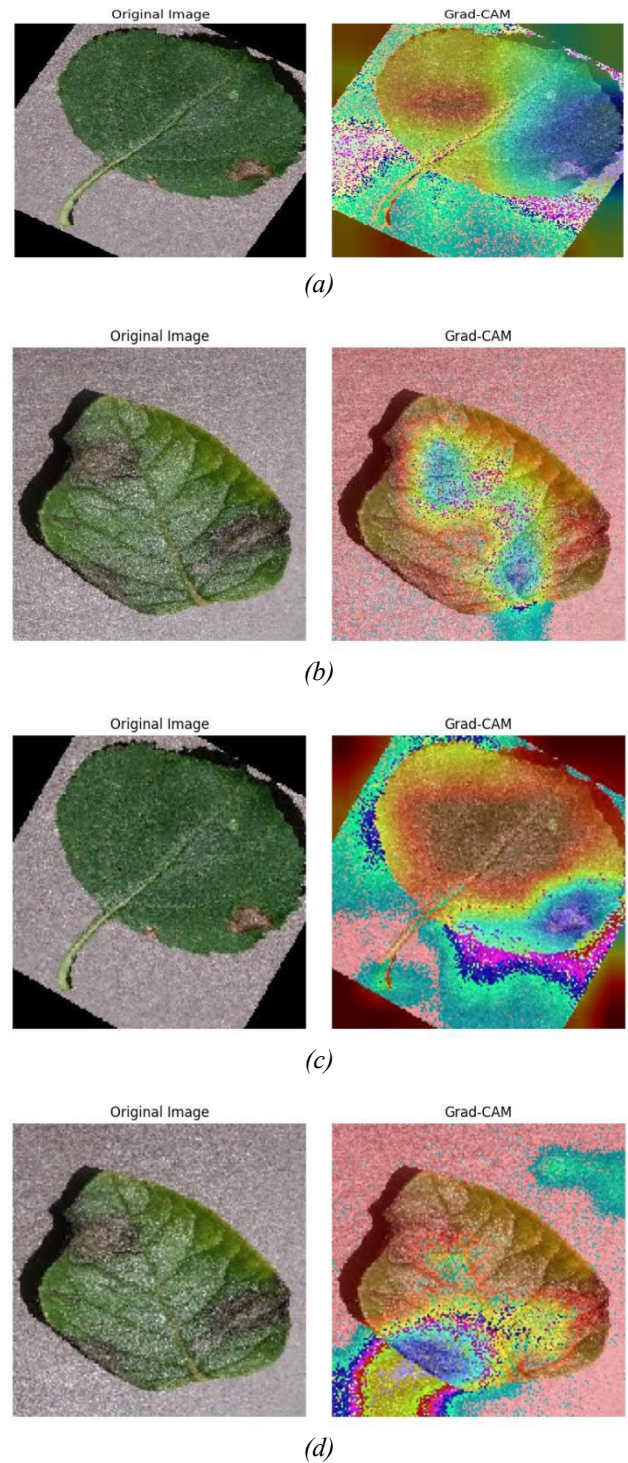


Fig. 15: Grad-CAM visualizations showing original images (left) and corresponding activation heatmaps (right) for different disease cases. (a) Apple Black Rot – MobileNetV2, (b) Potato Late Blight – MobileNetV2, (c) Apple Black Rot – CNN, (d) Potato Late Blight – CNN. The red and yellow areas indicate higher model attention toward diseased regions.

V. CONCLUSION

In conclusion, this study aimed to develop an automated crop disease classification system using leaf images, evaluating both a custom CNN architecture and a MobileNetV2-based model. Through comprehensive preprocessing, augmentation, and training procedures, the MobileNetV2 model demonstrated superior performance with an overall accuracy of 96.93% and consistently high precision, recall, and F1-scores across 33 crop disease and healthy leaf categories. In contrast, the custom CNN model achieved 80.83% accuracy, with notable variance in class-wise predictions.

These results emphasize the effectiveness of lightweight deep learning architectures, particularly MobileNetV2, for practical agricultural diagnostics. While the custom model showed potential, its performance highlights the need for architectural refinement and better feature learning strategies. The promising outcomes of the MobileNetV2 model support its applicability in real-world scenarios, offering a viable solution for early disease detection and improved crop management. Future work may focus on integrating explainable AI techniques, expanding dataset diversity, and deploying the system as a mobile or web-based tool to aid farmers and agricultural professionals in real-time decision-making.

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